

WASP-43b

R-0381

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250328 · created 2026-07-18T19:59:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460762.7581 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1893 +/- 0.0097
Transit depth (Rp/Rs)^2	3.58 +/- 0.37 [%]
Orbital Inclination (inc)	83.98 +/- 1.31
Airmass coefficient 1 (a1)	1.268 +/- 0.015
Airmass coefficient 2 (a2)	-0.167 +/- 0.053
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	None
Optimal Aperture	3.79
Optimal Annulus	4.96
Transit Duration (day)	0.0575 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1893 $\pm$ 0.0097 vs 0.1594 (deviation 3.08 σ)
Duration fit vs published	0.0575 d vs 0.0512 d (deviation 2.08 σ)
Mid-time O-C	-3.9 min
Depth SNR	19.5
Residual scatter	1.19 %

Light curve

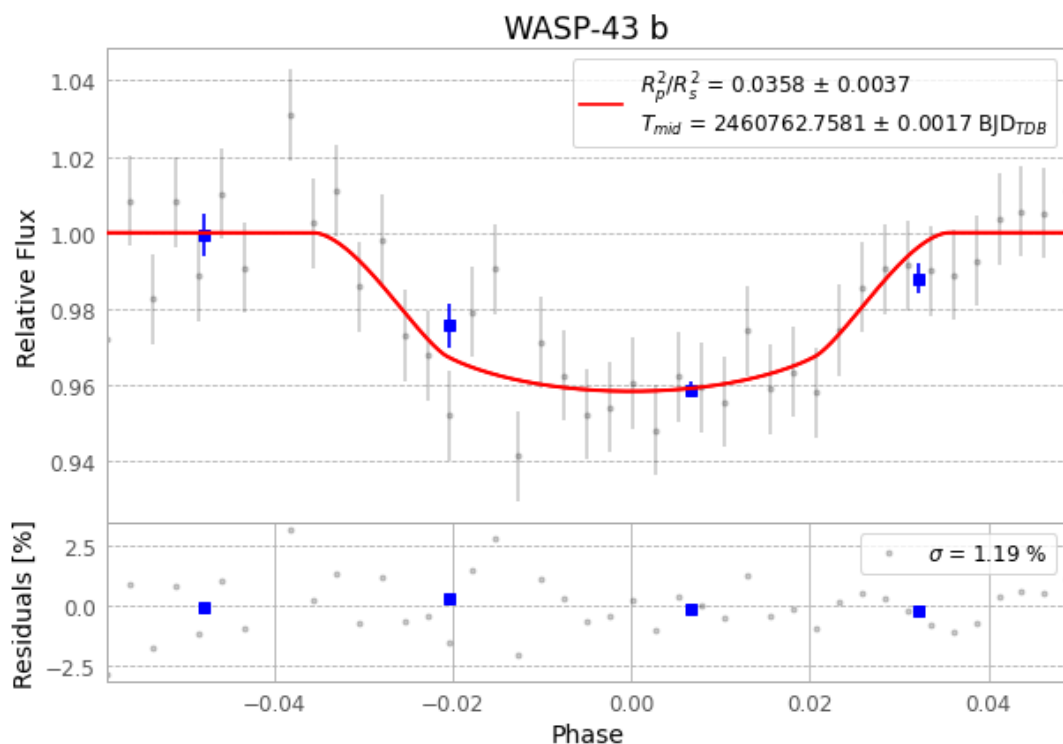


Figure 1 — FinalLightCurve_WASP-43 b_2025-03-27.png (SHA-256 3c411dea2025ad07...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [389, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5587e435b09f8b76...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

43 FITS frames (28 MB), WASP-43250328045415.FITS → WASP-43250328070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250328.log (38158032bc3c...), FinalLightCurve_WASP-43 b_2025-03-27.png (3c411dea2025...), FinalLightCurve_WASP-43 b_2025-03-27.csv (1024b04c20c9...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 19:59Z.